

Grade 10 ICT (Sri Lanka) – Operating Systems

What is an Operating System (OS)?

An **Operating System (OS)** is the most important system software in a computer. It acts as an **interface between the user and the computer hardware**, allowing users to run programs and manage computer resources efficiently.

Without an operating system, a computer cannot perform useful tasks for the user.

Main Functions of an Operating System

1. Process Management

The operating system manages all the programs running on the computer. The OS is responsible for:

- Starting programs.
- Stopping programs.
- Schedules CPU time.
- Allows multitasking (running several programs at once).

Example:

Listening to music while typing a document.

2. Memory Management

The OS controls the computer's main memory (RAM). The OS:

- Allocates memory to programs.
 - Prevents programs from interfering with each other.
 - Frees memory after programs close.
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3. File Management

The operating system organizes data into files and folders.

It allows users to:

- Create files
- Copy files
- Move files
- Rename files
- Delete files
- Search for files

Examples of file types:

- .docx
- .jpg
- .mp3
- .pdf

4. Device Management

The OS communicates with hardware devices through **device drivers**.

Devices include:

Keyboard 

Mouse 

Printer 

Scanner 

Monitor 

Webcam 

Speakers  

5. User Interface (UI)

The user interface allows people to communicate with the computer.

There are two common types.



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Graphical User Interface (GUI)

Uses:

- Windows
- Icons
- Menus
- Pointer (Mouse)

Examples:

- Microsoft Windows
- macOS
- Ubuntu Desktop

Advantages:

- Easy to learn
 - User friendly
 - Suitable for beginners
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Command Line Interface (CLI)

Users type commands using the keyboard.

Examples:

- Command Prompt
- Linux Terminal

Advantages:

- Faster for experienced users
- Uses fewer system resources

Disadvantages:

- Requires memorizing commands.
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Types of Operating Systems

Single User Operating System

Designed for one user at a time.

Examples include:

- Windows 10
 - Windows 11
 - macOS
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Multi-user Operating System

Allows many users to use the same computer simultaneously.

Examples:

- UNIX
 - Linux Servers
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Multitasking Operating System

Runs several applications simultaneously.

Example:

Typing a report while downloading files and playing music.

Real-Time Operating System (RTOS)

Responds immediately to events. Used in:

- Medical equipment
 - Aircraft systems
 - Industrial robots
 - Traffic control systems
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Booting

Booting is the process of starting a computer.

Cold Boot

Starting the computer from a powered-off state.

Warm Boot

Restarting the computer without switching off the power.

Device Drivers

A device driver is software that allows the operating system to communicate with hardware devices.

Without the correct driver:

- Printers may not work.
 - Graphics cards may not perform correctly.
 - Scanners may not be detected.
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Utility Software

Utility software helps maintain and protect the computer.

Examples include:

- Antivirus software
 - Disk Cleanup
 - Backup software
 - Disk Defragmenter (for HDDs)
 - File Compression software
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Popular Operating Systems

Operating System	Common Usage
• Microsoft Windows	• Personal computers
• Linux	• Servers and developers
• macOS	• Apple computers
• Android	• Smartphones and tablets
• iOS	• Apple iPhones and iPads

Advantages of an Operating System

- Makes computers easy to use.
 - Manages hardware efficiently.
 - Runs multiple applications.
 - Organizes files.
 - Improves security.
 - Controls computer resources.
 - Provides a user interface.
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Key Terms

- | | |
|----------------------|------------------------------|
| • Operating System | • CLI |
| • Hardware | • Device Driver |
| • Software | • Booting |
| • Process Management | • Utility Software |
| • Memory Management | • Multitasking |
| • File Management | • Single User |
| • Device Management | • Multi-user |
| • GUI | • Real-Time Operating System |



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Quick Revision Points

- An operating system is system software.
- It manages hardware and software resources.
- GUI uses windows, icons, menus, and a pointer.
- CLI requires typed commands.
- Device drivers connect hardware with the operating system.
- Utility software maintains and protects the computer.
- Booting is the process of starting the computer.
- Windows, Linux, macOS, Android, and iOS are popular operating systems.
- RAM is managed by the operating system.
- Files and folders are organized by the operating system.



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25 Practice Questions

Multiple Choice Questions

1. What is the primary purpose of an operating system?
 - a) Create websites
 - b) Manage computer hardware and software
 - c) Design graphics
 - d) Browse the Internet
2. Which of the following is **system software**?
 - a) Microsoft Word
 - b) Adobe Photoshop
 - c) Windows 11
 - d) VLC Media Player
3. Which operating system is mainly used on Apple Mac computers?
 - a) Android
 - b) Linux
 - c) macOS
 - d) iOS
4. Which operating system is commonly used on smartphones?
 - a) Windows Server
 - b) Android
 - c) Ubuntu Server
 - d) DOS
5. Which operating system function manages files and folders?
 - a) Process Management
 - b) File Management
 - c) Device Management
 - d) Memory Management
6. Which interface uses icons, windows and menus?
 - a) CLI
 - b) GUI
 - c) DOS
 - d) BIOS
7. Which interface requires typing commands?
 - a) GUI
 - b) Touch Interface
 - c) CLI
 - d) Desktop



8. What is a device driver?
 - a) A hardware device
 - b) Software that helps the OS communicate with hardware
 - c) A programming language
 - d) An antivirus program
9. Which of these is utility software?
 - a) Antivirus
 - b) PowerPoint
 - c) Chrome
 - d) Photoshop
10. What is a Cold Boot?
 - a) Restarting the computer
 - b) Starting from a powered-off state
 - c) Logging out
 - d) Sleep mode

Short Answer Questions

11. Define an Operating System.
12. State four functions of an Operating System.
13. What is memory management?
14. Explain file management.
15. What is multitasking?
16. Differentiate between GUI and CLI.
17. Give two examples of GUI operating systems.
18. What is the purpose of a device driver?
19. Explain the difference between Cold Boot and Warm Boot.
20. Name four utility software programs or categories.

Thinking / Application Questions

21. A school computer cannot print documents even though the printer is connected. What could be the reason?
22. Why is multitasking useful for a student doing research?
23. Why is an operating system necessary even if a computer has excellent hardware?
24. A hospital monitoring system must react immediately when a patient's heartbeat changes. Which type of operating system is most suitable? Explain your answer.
25. Explain how the operating system helps a user save, locate, and organize school assignments using files and folders.

